

CE 252: Geology and Geomorphology

Course Code	CE 252
Course Name	Geology and Geomorphology
L-T-P-C	2-0-2-3
Prerequisites	None
Intended for	UG
Distribution	Elective
Approval	23rd BoA (Apr 2018); 16th Senate (Apr 2018)

Course Contents

- **Module I**
Introduction, importance and significance of Earth Science, General characteristics and origin of the Universe, Solar System and its planets. [4 Lectures]
- **Module II**
Rock cycle, types and characteristics: Igneous, Sedimentary and Metamorphic rocks and their origin, texture, mineral composition, structure and classification. Rock properties. [5 Lectures]
- **Module III**
Plate tectonic theories, Physiographic sub divisions of India with focus on Himalayas. [4 Lectures]
- **Structural Geology**
Dip, strike, faults, folds, joints, emphasizing on Himalayan terrain. [6 Lectures]
- **Module V**
Basic concepts of Geomorphology, Cycle of erosion, Mountains and relief, river basin, drainage network, drainage types. [5 Lectures]
- **Module VI**
Geomorphic landforms, erosional & depositional: Fluvial, Glacial landforms. [4 Lectures]

Laboratory Modules

1. Identifying physical properties of rocks using visual interpretations. (4 hours)
2. Estimation of Hardness of rock using Mohr's Scale test. (2 hours)
3. Mineral identification using several properties. (2 hours)
4. Dip/ Strike measurements using Brunton compass in and around campus. (4 hours)
5. Geological map reading and interpretations. (4 hours)
6. Geomorphological mapping and interpretation of maps. (4 hours)
7. Geological field mapping around the campus. (8 hours)

Text Books

1. Lutgens & Tarbuck, **Essentials of Geology (with CD)** Pearson Education, 2012.
2. P. Singh, **Engineering and General Geology**, S. K. Kataria and Sons, 2009.

References

1. A L. Bloom, **Geomorphology: A systematic Analysis of Late Cenozoic Landforms**, 3rd Ed., Pearson Education, Inc., USA, 2004.
2. Peter MacLaren Donald Duff, Donald Duff, **Holme's Principles of physical geology**, Chapman & Hall, 1992.
3. F. G. Bell, **Engineering Geology**, Elsevier, 2007.
4. V S. Kale, and A. Gupta, **Introduction to Geomorphology**, Orient Longman Ltd., 2001.